8 Finance

8.2 Annuities

Formulas so far:

For these formulas, we have the following:

- P_0 is the pricipal.
- r is the interest rate.
- t is the number of time intervals (specified by problem) which have passed.
- A is the amount paid back.
- I is the interest.

Simple Interest	$I = P_0 r$ $A = P_0(1+r)$
Simple Interest over Time	$I = P_0 rt$ $A = P_0(1+rt)$

For these formulas, we have the following:

- P_0 is the pricipal.
- r is the annual interest rate.
- t is the number of years which have passed.
- A is the total amount.
- k is the number of times interest is compounded per year.

Compound Interest	$A = P_0\left(1+\frac{r}{k}\right)^{kt}$

Example 8.2.1 You will need \$40,000 for your childs education in 18 years. You open a new savings account with a 4% APR compounded quarterly. How much will you need to invest into the account now to have enough money when your child is 18?

Definition 8.2.2 (Savings Annuity) Placing a small amount in an account each month to slowly build up savings instead of placing a large sum to begin with.

Go over derivation:

1. DEFINE $r = APR$, k - number of compounds per year. d the deposit amount.
2. Write recursive formula for P_m, where m is the number of compounds that have occured.
3. Expand, and write two of these
4. multiply second one by $(1+r/k)$ then subtract to simplify.

Final formula $P_m = d\frac{\left(\left(1+\frac{r}{k}\right)^m - 1\right)}{\frac{r}{k}}$ one last step, $m = tk$, where t is the number of years that have passed.

Note: in this formula we assume that we make a deposit once every time the interest is compounded, so if there is a problem where the number of compounds per year is not stated, just use the number of deposits per year for k.

Example 8.2.3 Suppose you create an individual retirement account (IRA) with a 6% APR which is compounded monthly. Each month you deposit \$100 into the account. How much will be in the account in 20 years?

Example 8.2.4 You want to have \$200,000 in your account when you retire in 30 years. Your retirement account earns 8% interest. How much do you need to deposit each month to meet your retirement goal?

Example 8.2.5 You open a new bank account with a 3% APR, compounded daily, and each day you deposit \$10 into it. How much will you have in the account in 10 years?

Example 8.2.6 You want to have \$10,000 dollars saved in a special savings account for a trip in 3 years. The account has a 2.5% APR compounded twice a month. If you plan on depositing money into the account twice a month, how much will you need to deposit to reach your goal?